

Supplementary Topic: Discretizing AR(1) Processes

Tauchen (1986) and Rouwenhorst (1995)

Yasuyuki Sawada, Yaolang Zhong

University of Tokyo

yaolang.zhong@e.u-tokyo.ac.jp

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Motivation: Why Discretize?

Many economic models feature continuous stochastic processes:

$$z_{t+1} = \rho z_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2)$$

Examples:

- ▶ Log income/productivity shocks in consumption-saving models
- ▶ TFP shocks in firm dynamics models
- ▶ Idiosyncratic labor income in heterogeneous agent models

Problem: Solving the Bellman equation requires computing:

$$\mathbb{E}[V(a', z') \mid z] = \int V(a', z') f(z' \mid z) dz'$$

Solution: Approximate the continuous process with a discrete Markov chain.

Discretization vs. Quadrature

Two approaches to handle $\mathbb{E}[V(z') | z]$:

Quadrature (Lecture 7):

- ▶ Approximate $\mathbb{E}[V(z')|z]$ by $\sum_k w_k V(\rho z + \varepsilon_k)$.
- ▶ Nodes $\rho z + \varepsilon_k$ **move with** the current state z .
- ▶ Recompute nodes each time you evaluate the expectation.

Discretization (This Topic):

- ▶ Fix a global grid $\mathcal{Z} = \{z_1, \dots, z_N\}$ once.
- ▶ Build a transition matrix Π where $\Pi_{ij} = P(z' = z_j | z = z_i)$.
- ▶ Expectation becomes matrix-vector multiplication: $\mathbb{E}[V(z')|z_i] = \sum_j \Pi_{ij} V(z_j)$.

Key difference: Discretization is a **preprocessing step**; quadrature is done **inside the loop**.

The Setup: AR(1) Process

We focus on the standard AR(1) process:

$$z_{t+1} = \rho z_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2)$$

Key properties:

- ▶ Unconditional mean: $\mathbb{E}[z] = 0$ (assuming mean-zero process)
- ▶ Unconditional variance: $\sigma_z^2 = \frac{\sigma_\varepsilon^2}{1-\rho^2}$
- ▶ Autocorrelation: $\text{Corr}(z_t, z_{t+1}) = \rho$
- ▶ Conditional distribution: $z' \mid z \sim N(\rho z, \sigma_\varepsilon^2)$

Goal: Construct a discrete Markov chain $(z_1, \dots, z_N, \Pi)$ that approximates these properties.

Two main methods:

- ▶ Tauchen (1986): Integrate conditional density over bins
- ▶ Rouwenhorst (1995): Match moments exactly via clever construction

Tauchen's Method (1986): The Idea

Step 1: Choose the grid $\mathcal{Z} = \{z_1, \dots, z_N\}$

- ▶ Span the unconditional distribution: $z_1 = -m\sigma_z$, $z_N = +m\sigma_z$ (typically $m = 3$)
- ▶ Equal spacing: $z_j = z_1 + (j - 1) \cdot \Delta$, where $\Delta = \frac{z_N - z_1}{N - 1}$

Step 2: Define bins around each grid point

- ▶ Interior points: bin j is $[z_j - \Delta/2, z_j + \Delta/2]$
- ▶ Boundary points: extend to $\pm\infty$

Step 3: Compute transition probabilities

- ▶ Π_{ij} = probability that z' falls in bin j , given $z = z_i$
- ▶ Use the conditional distribution: $z' \mid z_i \sim N(\rho z_i, \sigma_\varepsilon^2)$

Tauchen's Method: The Formula

The transition probability from z_i to bin j :

For interior points ($1 < j < N$):

$$\Pi_{ij} = \Phi\left(\frac{z_j + \Delta/2 - \rho z_i}{\sigma_\varepsilon}\right) - \Phi\left(\frac{z_j - \Delta/2 - \rho z_i}{\sigma_\varepsilon}\right)$$

For boundary points:

$$\Pi_{i1} = \Phi\left(\frac{z_1 + \Delta/2 - \rho z_i}{\sigma_\varepsilon}\right), \quad \Pi_{iN} = 1 - \Phi\left(\frac{z_N - \Delta/2 - \rho z_i}{\sigma_\varepsilon}\right)$$

where $\Phi(\cdot)$ is the standard normal CDF.

Interpretation: We're computing the probability mass that lands in each bin, given the conditional normal distribution centered at ρz_i .

Tauchen's Method: Pros and Cons

Advantages:

- ▶ Simple and intuitive—just “bin the probability mass”
- ▶ Easy to implement (only needs the normal CDF)
- ▶ Generalizable to non-Gaussian shocks (if you have the CDF)
- ▶ Can be extended to VAR processes

Disadvantages:

- ▶ Struggles with high persistence ($\rho \rightarrow 1$)
 - ▶ When ρ is high, the conditional distribution $N(\rho z_i, \sigma_\varepsilon^2)$ is a narrow spike
 - ▶ Fixed grid may miss the spike or fail to capture the mass accurately
 - ▶ Approximated variance can be significantly off
- ▶ Requires many grid points for accurate approximation when $\rho > 0.9$

Rouwenhorst's Method (1995): The Idea

Key insight: Instead of integrating density areas, construct a Markov chain that **matches moments exactly**.

The method matches:

- ▶ Mean: $\mathbb{E}[z] = 0$ (by symmetric grid construction)
- ▶ Variance: $\text{Var}(z) = \sigma_z^2$ (by grid spacing)
- ▶ Autocorrelation: $\text{Corr}(z_t, z_{t+1}) = \rho$ (by transition matrix construction)

Building block: Start with a simple 2-state chain, then recursively build larger chains.

Intuition: The process z can be thought of as a sum of independent Bernoulli shocks. As $N \rightarrow \infty$, the stationary distribution (Binomial) approximates the Normal by CLT.

Rouwenhorst: Step 1 — The Grid

The grid is symmetric around zero:

$$z_j = -\psi + \frac{2\psi(j-1)}{N-1}, \quad j = 1, \dots, N$$

where $\psi = \sqrt{N-1} \cdot \sigma_z = \sqrt{N-1} \cdot \frac{\sigma_\epsilon}{\sqrt{1-\rho^2}}$.

So: $z_1 = -\psi$, $z_N = +\psi$, equally spaced.

Why this spacing?

- ▶ The stationary distribution of a Rouwenhorst chain is Binomial
- ▶ Binomial($N-1, 0.5$) has variance $\frac{N-1}{4}$ (in terms of “steps”)
- ▶ Scaling by $\frac{2\psi}{N-1}$ gives variance σ_z^2

Rouwenhorst: Step 2 — Base Case ($N = 2$)

For a 2-state chain, the transition matrix is:

$$\Pi_2 = \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}$$

What autocorrelation does this produce?

- ▶ States: $z_1 = -\psi$, $z_2 = +\psi$
- ▶ Stationary distribution: $(0.5, 0.5)$ (symmetric)
- ▶ Autocorrelation: $\text{Corr}(z_t, z_{t+1}) = 2p - 1$

To match the AR(1) autocorrelation ρ :

$$p = \frac{1 + \rho}{2}$$

Example: If $\rho = 0.9$, then $p = 0.95$ (high persistence = high probability of staying).

Rouwenhorst: Step 3 — Recursive Construction

Given Π_{N-1} , build Π_N by:

$$\Pi_N = p \begin{bmatrix} \Pi_{N-1} & \mathbf{0} \\ \mathbf{0}^\top & 0 \end{bmatrix} + (1-p) \begin{bmatrix} \mathbf{0} & \Pi_{N-1} \\ 0 & \mathbf{0}^\top \end{bmatrix} + (1-p) \begin{bmatrix} 0 & \mathbf{0}^\top \\ \Pi_{N-1} & \mathbf{0} \end{bmatrix} + p \begin{bmatrix} \mathbf{0}^\top & 0 \\ \mathbf{0} & \Pi_{N-1} \end{bmatrix}$$

Then normalize all rows except the first and last to sum to 1.

Intuition:

- ▶ Think of z as a sum of $N - 1$ independent Bernoulli random variables
- ▶ Each recursion adds one more Bernoulli shock
- ▶ By CLT: sum of Bernoullis $\rightarrow$ Normal distribution

Rouwenhorst: Pros and Cons

Advantages:

- ▶ Matches moments (mean, variance, autocorrelation) **exactly**
- ▶ Works well for **any** level of persistence, including $\rho \rightarrow 1$
- ▶ Requires fewer grid points than Tauchen for same accuracy
- ▶ Simple recursive construction

Disadvantages:

- ▶ Specifically designed for AR(1) with Gaussian innovations
- ▶ Hard to extend to:
 - ▶ Non-Gaussian shocks
 - ▶ VAR processes (multivariate)
 - ▶ Non-linear dynamics
- ▶ The construction is less intuitive than Tauchen

Tauchen vs. Rouwenhorst: Summary

	Tauchen	Rouwenhorst
Approach	Integrate density over bins	Match moments exactly
AR(1) Gaussian	✓	✓
High persistence ($\rho > 0.9$)	Poor	Excellent
Non-Gaussian shocks	✓ (if CDF known)	×
VAR extension	✓	×
Simplicity	Very simple	Simple (recursive)
Grid points needed	More	Fewer

Rule of thumb:

- ▶ AR(1) with $\rho < 0.9$: Either method works; Tauchen is simpler
- ▶ AR(1) with $\rho \geq 0.9$: Use Rouwenhorst
- ▶ Non-AR(1) or non-Gaussian: Use Tauchen (or simulation-based methods)

Practical Implementation

Both methods are implemented in standard libraries:

Python:

- ▶ `quantecon.markov.approximation.tauchen()`
- ▶ `quantecon.markov.approximation.rouwenhorst()`

Julia:

- ▶ `QuantEcon.tauchen()`
- ▶ `QuantEcon.rouwenhorst()`

Matlab:

- ▶ Various community implementations (e.g., on GitHub)

Typical usage:

Beyond AR(1): Brief Overview

What if your process is more complex?

AR(p) processes:

- ▶ Rewrite as VAR(1) in companion form, then use Tauchen
- ▶ Or approximate by AR(1) (often “good enough”)

Non-Gaussian shocks:

- ▶ Tauchen works if you can compute the conditional CDF
- ▶ Simulation-based methods: estimate Π from simulated paths

Modern alternatives:

- ▶ Farmer & Toda (2017): Maximum entropy discretization
- ▶ More general, can handle various distributions

References

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